

Phase 2: Reconnaissance Report

Target: 139.59.2.14

Assessment Phase: Vulnerability Assessment and Penetration Testing (VAPT)

Report Type: Comprehensive Reconnaissance Analysis

Date:

Team Members and Access Details:

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- Public IP: 111.92.61.232

- Tools Used: IPVoid, VirusTotal, Shodan, Censys

2. Team Member 2: Vaishnav K.R

- Public IP: 106.76.186.122

- Tools Used: Gobuster, Nikto, Whatweb

3. Team Member 1: Aswin Suresh

- Public IP: 202.83.36.228

- Tools Used: Nmap, Netcat(nc), Dig, Dnsrecon

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Authorization

Reconnaissance activities against IP 139.59.2.14 were conducted under course authorization from ICT Academy/Project Mentor - Mr. Hareesh Rajendran. Testing was limited to passive OSINT and low-impact active discovery; no exploitation, password guessing, or destructive testing was performed. Any sensitive data encountered was documented and not used beyond reporting.

Executive Summary

This comprehensive reconnaissance report presents detailed findings from both passive and active information gathering conducted against target IP 139.59.2.14. The assessment successfully identified a multi-service Ubuntu Linux server hosted on DigitalOcean infrastructure in Bangalore, India, with significant attack surface exposure through 14 open services including web applications, database systems, file sharing protocols, and directory services.

Key Findings:

- **14 open TCP ports** identified with critical services exposed
- **Multiple web applications** including phpMyAdmin administrative interface
- **Database services** with MariaDB instance accessible
- **Network file sharing** via NFS protocol
- **Directory services** through LDAP implementation
- **Outdated software components** with known security implications

1. Summary

1.1 Reconnaissance Scope and Objectives

Primary Objectives:

- Enumerate all accessible network services and ports
- Identify web applications and directory structures
- Determine operating system and software versions
- Map network topology and hosting infrastructure
- Assess security posture through passive analysis

Scope Definition:

- ★ **Target IP:** 139.59.2.14
- ★ **Network Range:** 139.59.2.0/24 (passive enumeration)
- ★ **Protocol Coverage:** TCP full port scan (1-65535)
- ★ **Service Analysis:** HTTP, FTP, SSH, DNS, LDAP, NFS, RPC, MySQL

1.2 Information Gathering Methodology

Passive Reconnaissance Tools:

- DNS reverse lookup and zone enumeration
- WHOIS registration analysis
- Network range reconnaissance
- Certificate transparency analysis

Active Reconnaissance Tools:

- Nmap comprehensive service enumeration
- Web application directory discovery
- Banner grabbing and service fingerprinting
- Vulnerability scanning with Nikto

1.3 Key Intelligence Findings

Infrastructure Profile:

- **Cloud Provider:** DigitalOcean, LLC
- **Geographic Location:** Bangalore, India (BLR1 datacenter)
- **Server Hostname:** ubuntu-s-1vcpu-1gb-blr1-cyber-03-10-2024-s-1vcpu-1gb-blr1-01
- **Operating System:** Ubuntu Linux 20.04
- **Network Allocation:** 139.59.0.0/16 (APNIC managed)

Critical Service Exposure:

- Multiple web servers on different ports (80, 54444, 56686)
- Database administration interface (phpMyAdmin)
- Network file sharing protocols (NFS)
- Directory services (LDAP)
- Remote access services (SSH, FTP)

1.4 Identified Attack Vectors and Entry Points

High-Priority Attack Vectors:

1. **Web Application Exploitation** - phpMyAdmin interface
2. **Database Direct Access** - MariaDB on port 53346
3. **File System Access** - NFS mounts and FTP services
4. **Directory Enumeration** - LDAP service exposure

5. Service Version Exploits - Outdated Apache/PHP components

Potential Entry Points:

- HTTP/HTTPS web applications (ports 80, 54444, 56686)
- FTP service with potential anonymous access
- SSH service for credential-based attacks
- Database services for injection attacks
- NFS shares for file system access

2. Technical Findings

2.1 Target System Architecture

System Identification:

```
Host:
ubuntu-s-1vcpu-1gb-blrl-cyber-03-10-2024-s-1vcpu-1gb-blrl-01 OS:
Ubuntu Linux 20.04 LTS
Architecture: x86_64 (1 vCPU, 1GB RAM
configuration) Virtualization: DigitalOcean
Droplet
```

Operating System Detection Results:

```
OS CPE: cpe:/o:linux:linux_kernel
Service Info: Host:
ubuntu-s-1vcpu-1gb-blrl-cyber-03-10-2024-s-1vcpu-1gb-blrl-01 OS Confidence:
Linux kernel detection confirmed via service banners
```

2.2 Network Topology and Service Enumeration

Complete Port and Service Analysis:

Port	Protocol	Service	Version	Status
21/tcp	FTP	OpenBSD ftpd	6.4 (Linux port 0.17)	Open
22/tcp	SSH	OpenSSH	8.2p1 Ubuntu 4ubuntu0.13	Open
53/tcp	DNS	ISC BIND	9.18.30-0ubuntu0.20.04.2	Open
80/tcp	HTTP	Apache httpd	2.4.41 (Ubuntu)	Open
111/tcp	RPC	rpcbind	2-4 (RPC #100000)	Open
389/tcp	LDAP	OpenLDAP	2.2.X - 2.3.X	Open
2049/tcp	NFS	NFS	3-4 (RPC #100003)	Open

39533/tcp	RPC	mountd	1-3 (RPC #100005)	Open
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Port	Protocol	Service	Version	Status
41661/tcp	RPC	nlockmgr	1-4 (RPC #100021)	Open
44251/tcp	RPC	mountd	1-3 (RPC #100005)	Open
53346/tcp	MySQL	MariaDB	10.3.23 or earlier	Open
54444/tcp	HTTP	Apache httpd	2.4.38 (Unix) OpenSSL/1.0.2q PHP/5.6.40	Open
55039/tcp	RPC	mountd	1-3 (RPC #100005)	Open
56686/tcp	HTTP	Apache httpd	2.4.38 (Unix) OpenSSL/1.0.2q PHP/5.6.40	Open

Network Topology Analysis:

```
Traceroute Results:
1 139.59.2.14 (139.59.2.14) 70.062 ms 50.324 ms 67.263 ms

Network Distance: 1 hop (direct routing)
Latency Range: 25-70ms (consistent with India-based hosting)
```

2.3 Application Mapping and Functionality Analysis

Web Application Discovery:

Primary Web Server (Port

```
Server: Apache/2.4.41 (Ubuntu)
Title: Apache2 Ubuntu Default Page: It works
Directory Structure:
├─/index.html (Status: 200) [Size: 10918]
├─/javascript/ (Status: 301) [Redirect]
├─/phpmyadmin/ (Status: 301) [Redirect]
└─/server-status (Status: 403) [Forbidden]
```

Secondary Web Applications:

Port 54444 Analysis:

```
Server: Apache/2.4.38 (Unix) OpenSSL/1.0.2q PHP/5.6.40 mod_perl/2.0.8-dev Perl/v5.16.3
Title: Bad request!
SSL Certificate: localhost/Apache Friends/Berlin/DE (Expired: 2010-09-30)
```

Port 56686 Analysis:

```
Server: Apache/2.4.38 (Unix) OpenSSL/1.0.2q PHP/5.6.40 mod_perl/2.0.8-dev Perl/v5.16.3
Title: ICTAK Vulnerable Web Application : The World of Blogging
Robots.txt: Disallows all paths (/)
```

Database Services:

```
MariaDB Instance (Port 53346):
Version: 10.3.23 or earlier
Status: Unauthorized access (authentication required)
```

Directory Services (LDAP - Port 389):

```
LDAP Root Context: dc=nodomain
Administrative Account:
cn=admin,dc=nodomain Directory Structure:
├── dc=nodomain (organization object)
└── cn=admin,dc=nodomain (administrative role)
```

Network File System (NFS - Port 2049):

```
NFS Version: 3-4 (RPC #100003)
Mount Daemon Ports: 39533, 44251, 55039
File Locking Service: 41661 (nlockmgr)
Status: No exported mounts visible without authentication
```

2.4 Banner Grabbing Evidence**FTP Service Banner:**

```
220 ubuntu-s-1vcpu-1gb-blrl-cyber-03-10-2024-s-1vcpu-1gb-blrl-01
FTP server (Version 6.4/OpenBSD/Linux-ftpd-0.17) ready.
```

SSH Service Banner:

```
SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.13
```

HTTP Service Response:

```
HTTP/1.1 400 Bad Request
Date: Tue, 07 Oct 2025 10:03:56 GMT
Server: Apache/2.4.41
(Ubuntu) Content-Length: 301
Connection: close
Content-Type: text/html; charset=iso-8859-1
```

2.5 Infrastructure Registration Analysis**WHOIS Information:**

```

Network Range: 139.59.0.0 - 139.59.15.255
Registry: APNIC (Asia Pacific Network Information Centre)
Organization: DigitalOcean, LLC

```

```

Country: IN (India)
Abuse Contact:
abuse@digitalocean.com
Administrative Contact: DOIA2-AP
Last Modified: 2020-05-31T21:32:24Z

```

Network Neighbors Analysis:

46 active hosts identified in 139.59.2.0/24 subnet, including:

- Multiple MongoDB databases (staging and production)
- CloudWays applications
- Business domains and services
- Development/testing environments

3. Intelligence Assessment

3.1 Information Sensitivity Analysis

High Sensitivity Exposures:

- **phpMyAdmin Interface:** Direct database administration access
- **LDAP Directory Services:** Organizational structure and user accounts
- **NFS File Shares:** Potential access to file system contents
- **Database Services:** Direct MariaDB access on non-standard port

Medium Sensitivity Exposures:

- **Web Applications:** Multiple applications with varying security postures
- **Service Banners:** Detailed version information enabling targeted attacks
- **DNS Services:** Internal hostname and infrastructure details

Low Sensitivity Exposures:

- **SSH Service:** Standard remote access (requires credentials)
- **HTTP Default Pages:** Minimal information disclosure

3.2 Exposure Risk Evaluation

Critical Risk Factors:

1. **Multiple Web Applications** with different security configurations
2. **Outdated Software Components** (Apache 2.4.38, PHP 5.6.40)

3. **Database Administrative Interface** accessible via web
4. **Network File Sharing** protocols exposed to internet
5. **Directory Services** with default configuration

Risk Severity Scoring:

- **Critical (9-10):** phpMyAdmin exposure, outdated PHP version
- **High (7-8):** MariaDB direct access, NFS services
- **Medium (4-6):** LDAP services, multiple HTTP services
- **Low (1-3):** SSH service, DNS services

3.3 Attack Surface Quantification

Service Distribution:

- **Web Services:** 3 instances (ports 80, 54444, 56686)
- **Database Services:** 2 instances (MariaDB, phpMyAdmin)
- **File Services:** 2 protocols (FTP, NFS)
- **Directory Services:** 1 instance (LDAP)
- **Remote Access:** 2 services (SSH, FTP)
- **Network Services:** 2 instances (DNS, RPC)

Vulnerability Surface Metrics:

- **Total Open Ports:** 14
- **Web Attack Vectors:** 5+ (applications, directories, forms)
- **Network Protocols:** 8+ different service types
- **Authentication Points:** 6 services requiring credentials

3.4 Threat Landscape Contextualization

Current Threat Environment:

- **Web Application Attacks:** High prevalence against phpMyAdmin
- **Database Exploitation:** Common targeting of exposed MySQL/MariaDB
- **Cloud Infrastructure:** Increased scanning of DigitalOcean ranges
- **Automated Scanning:** Regular reconnaissance against common ports

Industry-Specific Risks:

- **Educational Sector:** ICTAK application suggests academic target
- **Development Environment:** Multiple staging/testing indicators
- **Shared Hosting:** High-density subnet increases collateral risk

4. Security Findings Summary

4.1 Web Application Security Issues

Nikto Vulnerability Scan Results:

```

Target: 139.59.2.14:80
Scan Duration: 472 seconds (8102 requests)

Identified Issues:
• Missing X-Frame-Options header (clickjacking vulnerability)
• Missing X-Content-Type-Options header
• Apache 2.4.41 outdated (current version 2.4.54+)
• Server information disclosure via ETags (CVE-2003-1418)
• phpMyAdmin directory accessible
• Insecure cookie configuration (missing httponly flag)
• Allowed HTTP methods: POST, OPTIONS, HEAD, GET

```

4.2 Service Version Vulnerabilities

Outdated Components:

- **Apache 2.4.38:** Multiple known CVEs, end-of-life for security updates
- **PHP 5.6.40:** End-of-life since January 2017, no security updates
- **OpenSSL 1.0.2q:** Deprecated version with known vulnerabilities
- **MariaDB 10.3.23:** Several security advisories available

4.3 Configuration Security Issues

Insecure Configurations:

- Multiple web servers on non-standard ports
- Default Apache configuration pages exposed
- LDAP service with minimal access control
- NFS services accessible without proper authentication
- Expired SSL certificates (2010 expiration)

5. Recommendations

5.1 Immediate Actions Required

1. **Restrict phpMyAdmin Access:** Implement IP restrictions and strong authentication
2. **Update Software Components:** Upgrade Apache, PHP, and MariaDB to current versions
3. **Secure Web Applications:** Remove default pages and implement security headers
4. **Configure Firewall Rules:** Limit exposure of database and file sharing services
5. **Implement SSL/TLS:** Update certificates and enforce HTTPS

5.2 Medium-Term Security Improvements

1. **Network Segmentation:** Isolate database and file services from public internet
2. **Access Control Implementation:** Strengthen LDAP and NFS authentication
3. **Monitoring and Logging:** Implement comprehensive security monitoring
4. **Regular Security Assessments:** Establish ongoing vulnerability management
5. **Backup and Recovery:** Ensure secure backup procedures for critical data

6. Conclusion

The reconnaissance phase successfully identified a complex multi-service environment with significant security exposure. The target system presents multiple attack vectors through web applications, database services, and network file sharing protocols. The combination of outdated software components, insecure configurations, and broad service exposure creates a high-risk security posture requiring immediate attention.

The findings provide a comprehensive foundation for subsequent vulnerability assessment and penetration testing phases, with clear prioritization of critical security issues requiring remediation.

Appendix: Evidence Documentation

A.1 Command Execution Log

DNS Reconnaissance:

```
dig -x 139.59.2.14
dnsrecon -r 139.59.2.0/24
```

Network Scanning:

```
nmap -sS -sV -p 1-65535 139.59.2.14
nmap -O 139.59.2.14
nmap --script banner 139.59.2.14
```

Web Application Analysis:

```
gobuster dir -u http://139.59.2.14 -w /usr/share/wordlists/dirb/common.txt
nikto -h http://139.59.2.14
whatweb http://139.59.2.14
```

Banner Grabbing:

```
nc -nv 139.59.2.14 80
nc -nv 139.59.2.14 22
```

A.2 Complete Service Enumeration

1. dig -x 139.59.2.14

```
1 |
2 ; <<>> DiG 9.20.11-4+b1-Debian <<>> -x 139.59.2.14
3 ;; global options: +cmd
4 ;; Got answer:
5 ;; ->HEADER<- opcode: QUERY, status: NXDOMAIN, id: 5113
6 ;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1
7
8 ;; OPT PSEUDOSECTION:
9 ; EDNS: version: 0, flags:; udp: 512
0 ;; QUESTION SECTION:
1 ;14.2.59.139.in-addr.arpa.      IN      PTR
2
3 ;; AUTHORITY SECTION:
4 2.59.139.in-addr.arpa. 1727  IN      SOA      ns1.digitalocean.com. hostmaster.2.59.139.in-addr.arpa. 1759748995 10800 3600 604800 1800
5
6 ;; Query time: 352 msec
7 ;; SERVER: 10.0.2.3#53(10.0.2.3) (UDP)
8 ;; WHEN: Tue Oct 07 10:23:57 IST 2025
9 ;; MSG SIZE rcvd: 120
0
```

2. dnsrecon -r 139.59.2.14

Performing Reverse Lookup from 139.59.2.0 to 139.59.2.255

PTR bep-emis.edsglobal.com 139.59.2.3

PTR kda.orgmatix.com 139.59.2.19

PTR 906586.cloudwaysapps.com 139.59.2.22

PTR db-mongodb-blr1-staging-4e290b68.mongo.ondigitalocean.com 139.59.2.25

PTR cpanel-whm-for-Innikiind.com 139.59.2.27

PTR panel.onomise.com 139.59.2.34

PTR rajchandra.me 139.59.2.40

PTR dns.aipi.in 139.59.2.41

PTR in01.carbongames.com 139.59.2.47

PTR 1304475.cloudwaysapps.com 139.59.2.50

PTR namaste-uat-dd3871b4.mongo.ondigitalocean.com 139.59.2.69

PTR bangalore.wonderproxy.com 139.59.2.71

PTR frappe-erp.fixv.co 139.59.2.74

PTR nebula.baapstore.com 139.59.2.75

PTR vps.newgandhiroadlines.com 139.59.2.81

PTR gd-prod-lnx-202.hrstop.com 139.59.2.82

PTR bangalore2.ontear.net 139.59.2.86

PTR advance-mame.com 139.59.2.88

PTR kirk.wifiaccess.in 139.59.2.99

PTR gateway.basezap.com 139.59.2.102

PTR chatroom.nixtree.com 139.59.2.103

PTR 1363443.cloudwaysapps.com 139.59.2.106

PTR wow.interactivedns.com 139.59.2.108

PTR idmx-database-6c1a8b27.mongo.ondigitalocean.com 139.59.2.112

PTR 405574.cloudwaysapps.com 139.59.2.114

PTR 1449253.cloudwaysapps.com 139.59.2.120

PTR 1314523.cloudwaysapps.com 139.59.2.113

PTR in-01.miningrigrentals.com 139.59.2.141

PTR upscminds.mizobot.com 139.59.2.145

PTR eidesign.org-ftp 139.59.2.159

PTR wastickers.me 139.59.2.167

PTR shretheducation.com 139.59.2.169

PTR 802659.cloudwaysapps.com 139.59.2.176

PTR new-ludo-engine-database-242c1456.mongo.ondigitalocean.com 139.59.2.188

PTR mywonderteam.com 139.59.2.190

PTR db-mongodb-blr1-96889-f44638c1.mongo.ondigitalocean.com 139.59.2.202

PTR scale-test-1738964325084-8cee285f.mongo.ondigitalocean.com 139.59.2.203

PTR staging.ei 139.59.2.212

PTR batteryboss.eshostingservices.net 139.59.2.221

PTR blr1-target.thousandeyes.analogpuddle.net 139.59.2.223

PTR s6.appyogi.com 139.59.2.225

PTR sameekshas.com 139.59.2.227

PTR 462216.cloudwaysapps.com 139.59.2.224

PTR arcr.finance 139.59.2.240

PTR favivomedia.com 139.59.2.246

PTR smtpb2.falconide.com 139.59.2.252

46 Records Found

3. **nmap -sS -sV -p 1-65535 139.59.2.14**

Starting Nmap 7.95 (<https://nmap.org>) at 2025-10-07 12:32 IST

Stats: 0:01:28 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan

SYN Stealth Scan Timing: About 12.42% done; ETC: 12:44 (0:10:20 remaining)

Stats: 0:01:42 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan

SYN Stealth Scan Timing: About 14.76% done; ETC: 12:43 (0:09:49 remaining)

Stats: 0:06:14 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan

SYN Stealth Scan Timing: About 96.99% done; ETC: 12:38 (0:00:12 remaining)

Stats: 0:08:16 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan

SYN Stealth Scan Timing: About 99.99% done; ETC: 12:40 (0:00:00 remaining)

Nmap scan report for 139.59.2.14

Host is up (0.20s latency).

Not shown: 65370 closed tcp ports (reset), 151 filtered tcp ports (no-response)

Some closed ports may be reported as filtered due to --defeat-rst-ratelimit

PORT STATE SERVICE VERSION

21/tcp open ftp OpenBSD ftpd 6.4 (Linux port 0.17)

22/tcp open ssh OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.0)

53/tcp open domain ISC BIND 9.18.30-0ubuntu0.20.04.2 (Ubuntu Linux)

80/tcp open http Apache httpd 2.4.41 ((Ubuntu))

111/tcp open rpcbind 2-4 (RPC #100000)

389/tcp open ldap OpenLDAP 2.2.X - 2.3.X

2049/tcp open nfs 3-4 (RPC #100003)

39533/tcp open mountd 1-3 (RPC #100005)

41661/tcp open nlockmgr 1-4 (RPC #100021)

44251/tcp open mountd 1-3 (RPC #100005)

53346/tcp open mysql MariaDB 10.3.23 or earlier (unauthorized)

54444/tcp open http Apache httpd 2.4.38 ((Unix) OpenSSL/1.0.2q PHP/5.6.40 mod_perl/2.0.8-dev Perl/v5.16.3)

55039/tcp open mountd 1-3 (RPC #100005)

56686/tcp open http Apache httpd 2.4.38 ((Unix) OpenSSL/1.0.2q PHP/5.6.40 mod_perl/2.0.8-dev Perl/v5.16.3)

Service Info: Host: ubuntu-s-1vcpu-1gb-blr1-cyber-03-10-2024-s-1vcpu-1gb-blr1-01; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at <https://nmap.org/submit/>.

Nmap done: 1 IP address (1 host up) scanned in 665.67 seconds

4. nmap -O 139.59.2.14

Starting Nmap 7.95 (<https://nmap.org>) at 2025-10-07 11:54 IST

Nmap scan report for 139.59.2.14

Host is up (0.0073s latency).

Not shown: 993 filtered tcp ports (no-response)

PORT STATE SERVICE VERSION

21/tcp open ftp OpenBSD ftpd 6.4 (Linux port 0.17)

22/tcp open ssh OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.0)

| ssh-hostkey:

| 3072 f0:5f:7b:6d:90:42:92:00:4a:e0:2b:02:0e:f6:74:bc (RSA)

```

| 256 92:ce:8f:ef:f0:96:bd:dc:89:7a:b1:b7:0c:b0:e5:db (ECDSA)
|_ 256 f7:29:e3:d5:cb:9d:63:ae:08:1f:32:ed:55:b0:10:19 (ED25519)
53/tcp open domain ISC BIND 9.18.30-0ubuntu0.20.04.2 (Ubuntu Linux)
| dns-nsid:
|_ bind.version: 9.18.30-0ubuntu0.20.04.2-Ubuntu
80/tcp open http Apache httpd 2.4.41 ((Ubuntu))
|_http-server-header: Apache/2.4.41 (Ubuntu)
|_http-title: Apache2 Ubuntu Default Page: It works
111/tcp open rpcbind 2-4 (RPC #100000)
| rpcinfo:
| program version port/proto service
| 100000 2,3,4 111/tcp rpcbind
| 100000 2,3,4 111/udp rpcbind
| 100000 3,4 111/tcp6 rpcbind
| 100000 3,4 111/udp6 rpcbind
| 100003 3 2049/udp nfs
| 100003 3 2049/udp6 nfs
| 100003 3,4 2049/tcp nfs
| 100003 3,4 2049/tcp6 nfs
| 100005 1,2,3 34386/udp mountd
| 100005 1,2,3 38478/udp6 mountd
| 100005 1,2,3 53471/tcp6 mountd
| 100005 1,2,3 55039/tcp mountd
| 100021 1,3,4 38757/tcp6 nlockmgr
| 100021 1,3,4 41661/tcp nlockmgr
| 100021 1,3,4 43103/udp6 nlockmgr
| 100021 1,3,4 56736/udp nlockmgr
| 100227 3 2049/tcp nfs_acl
| 100227 3 2049/tcp6 nfs_acl
| 100227 3 2049/udp nfs_acl
|_ 100227 3 2049/udp6 nfs_acl
389/tcp open ldap OpenLDAP 2.2.X - 2.3.X

```


2049/tcp open nfs 3-4 (RPC #100003)

Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port

Device type: bridge|VoIP adapter|general purpose

Running (JUST GUESSING): Oracle Virtualbox (97%), Slirp (97%), AT&T embedded (95%), QEMU (93%)

OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu

Aggressive OS guesses: Oracle Virtualbox Slirp NAT bridge (97%), AT&T BGW210 voice gateway (95%), QEMU user mode network gateway (93%)

No exact OS matches for host (test conditions non-ideal).

Network Distance: 1 hop

Service Info: Host: ubuntu-s-1vcpu-1gb-blr1-cyber-03-10-2024-s-1vcpu-1gb-blr1-01; OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE (using port 80/tcp)

HOP RTT ADDRESS

1 0.29 ms 139.59.2.14

OS and Service detection performed. Please report any incorrect results at <https://nmap.org/submit/>.

Nmap done: 1 IP address (1 host up) scanned in 32.36 seconds

Starting Nmap 7.95 (<https://nmap.org>) at 2025-10-07 15:14 IST

5. nmap - - script banner 139.59.2.14

nmap --script banner 139.59.2.14

Starting Nmap 7.95 (<https://nmap.org>) at 2025-10-07 15:51 IST

Nmap scan report for 139.59.2.14

Host is up (0.036s latency).

Not shown: 993 filtered tcp ports (no-response)

PORT STATE SERVICE

21/tcp open ftp

| banner: 220 ubuntu-s-1vcpu-1gb-blr1-cyber-03-10-2024-s-1vcpu-1gb-blr1-0

|_1 FTP server (Version 6.4/OpenBSD/Linux-ftpd-0.17) ready.

22/tcp open ssh

|_banner: SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.13

53/tcp open domain

80/tcp open http

111/tcp open rpcbind

389/tcp open ldap

2049/tcp open nfs

Nmap done: 1 IP address (1 host up) scanned in 21.74 seconds

6. gobuster dir -u http://139.59.2.14 -w /usr/share/wordlists/dirb/common.txt

```

1 =====
2 Gobuster v3.8
3 by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
4 =====
5 [+] Url: http://139.59.2.14
6 [+] Method: GET
7 [+] Threads: 10
8 [+] Wordlist: /usr/share/wordlists/dirb/common.txt
9 [+] Negative Status codes: 404
10 [+] User Agent: gobuster/3.8
11 [+] Timeout: 10s
12 =====
13 Starting gobuster in directory enumeration mode
14 =====
15 /.htpasswd (Status: 403) [Size: 276]
16 /.htaccess (Status: 403) [Size: 276]
17 /.hta (Status: 403) [Size: 276]
18 /index.html (Status: 200) [Size: 10918]
19 /javascript (Status: 301) [Size: 315] [→ http://139.59.2.14/javascript/]
20 /phpmyadmin (Status: 301) [Size: 315] [→ http://139.59.2.14/phpmyadmin/]
21 /server-status (Status: 403) [Size: 276]
22 =====
23 Finished
24 =====
25

```

7. nikto -h http://139.59.2.14

```

1- Nikto v2.5.0
2
3+ Target IP:      139.59.2.14
4+ Target Hostname: 139.59.2.14
5+ Target Port:    80
6+ Start Time:     2025-10-07 15:53:18 (GMT5.5)
7
8+ Server: Apache/2.4.41 (Ubuntu)
9+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
10+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
11+ No CGI Directories found (use '-C all' to force check all possible dirs)
12+ Apache/2.4.41 appears to be outdated (current is at least Apache/2.4.54). Apache 2.2.34 is the EOL for the 2.x branch.
13+ /: Server may leak inodes via ETags, header found with file /, inode: 2aa6, size: 5e3c7f0be936f, mtime: gzip. See: http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2003-1418
14+ OPTIONS: Allowed HTTP Methods: POST, OPTIONS, HEAD, GET .
15+ /phpmyadmin/changelog.php: Uncommon header 'x-ob_mode' found, with contents: 1.
16+ /phpmyadmin/changelog.php: Cookie goto created without the httponly flag. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Cookies
17+ /phpmyadmin/changelog.php: Cookie back created without the httponly flag. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Cookies
18+ /phpmyadmin/: phpMyAdmin directory found.
19+ 8102 requests: 0 error(s) and 9 item(s) reported on remote host
20+ End Time:       2025-10-07 16:01:10 (GMT5.5) (472 seconds)
21
22+ 1 host(s) tested
23

```

8. whatweb http://139.59.2.14

```

(aswin@kalilinux)-[~/Documents/VAPT]
$ cat whatweb.txt
http://139.59.2.14 [200 OK] Apache[2.4.41], Country[AUSTRALIA][AU], HTTPServer[Ubuntu Linux][Apache/2.4.41 (Ubuntu)], IP[139.59.2.14], Title[Apache2 Ubuntu Default Page: It works]
(aswin@kalilinux)-[~/Documents/VAPT]
$

```

9. nc -nv 139.59.2.14 80

```

1 nc -nv 139.59.2.14 80
2
3 (UNKNOWN) [139.59.2.14] 80 (http) open
4 GET / HTTP/1.1
5 Host: 139.59.2.14
6 Connection: close
7 HTTP/1.1 400 Bad Request
8 Date: Tue, 07 Oct 2025 10:03:56 GMT
9 Server: Apache/2.4.41 (Ubuntu)
10 Content-Length: 301
11 Connection: close
12 Content-Type: text/html; charset=iso-8859-1
13
14 <!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
15 <html><head>
16 <title>400 Bad Request</title>
17 </head><body>
18 <h1>Bad Request</h1>
19 <p>Your browser sent a request that this server could not understand.<br />
20 </p>
21 <hr>
22 <address>Apache/2.4.41 (Ubuntu) Server at 127.0.1.1 Port 80</address>
23 </body></html>
24

```

10. nc -nv 139.59.2.14 22

```
26 nc -nv 139.59.2.14 22
27 |
28 (UNKNOWN) [139.59.2.14] 22 (ssh) open
29 SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.13
```

A.3 Online Intelligence Platforms

1. WHOIS and IP Registration Analysis:
 - IPVoid WHOIS Lookup: <https://www.ipvoid.com/whois/>
 - Purpose: Domain registration details, ownership information, and contact data
2. Threat Intelligence and Reputation Analysis:
 - VirusTotal IP Analysis: <https://www.virustotal.com/gui/ip-address/139.59.2.14>
 - Purpose: Malware detection, URL scanning, and security vendor assessments
3. Internet Asset Discovery:
 - Shodan Search Engine: <https://www.shodan.io/host/139.59.2.14>
 - Purpose: Internet-connected device discovery and service enumeration
4. Certificate and Asset Intelligence:
 - Censys Internet Search: <https://platform.censys.io/hosts/139.59.2.14>
 - Purpose: Certificate transparency logs and internet-wide scanning data

References

1. IPVoid WHOIS Lookup: <https://www.ipvoid.com/whois/>
2. VirusTotal IP Analysis: <https://www.virustotal.com/gui/home/search>
3. Shodan Search Engine: <https://www.shodan.io/>
4. Censys Internet Search: <https://platform.censys.io/home>
5. Perplexity: <https://www.perplexity.ai/>